Class test - 5

Need to prove that

$$L = \{a^n b^n \mid n \geq 0\}$$

is not regular

using pumping lemma

Assuming the opposite that

L is regular

According to pumping lemma

p is the pumping length

$w \in L$

p can be divided into 3

$$w = xyz$$

$\therefore |xy| \leq p$

$|y| > 0$

$xy^i z \in L \quad i \geq 0$

Choosing $w = a^p b^p$

$w \in L$

$|xy| \leq p$, both x, y

contains a's

$y = a^k, \quad k \geq 1$

Pump the string

$i = 0$

$xy^i z = xy^0 z$

$xz = a^{p-k} b^p$

Since the powers are not equal

$a^{p-k} b^p \notin L$

Hence L is not regular because pumped string must belong to L

Comparative study of finite automata, pushdown automata and Turing machine

No.	F. A	PDA	T. M
Memory	No extra memory	uses a stack	uses an infinite tape
Storage	only finite states	stack follows LIFO order	Tape can read and write data
Language	Regular languages	context-free language	Recursively enumerable language
Power	Least powerful	more powerful than FA	most powerful of three